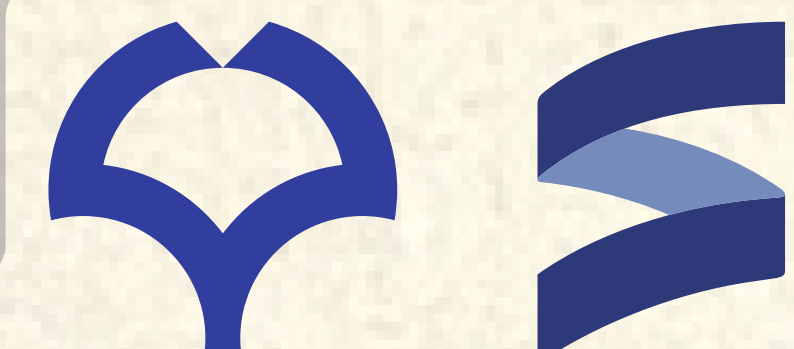




奥山に
紅葉踏み分け
鳴く鹿の
声聞くときぞ
秋は悲しき

Deep mountains
red leaves-treading and parting
crying deer
voice/sound hearing time
autumn is sorrowful

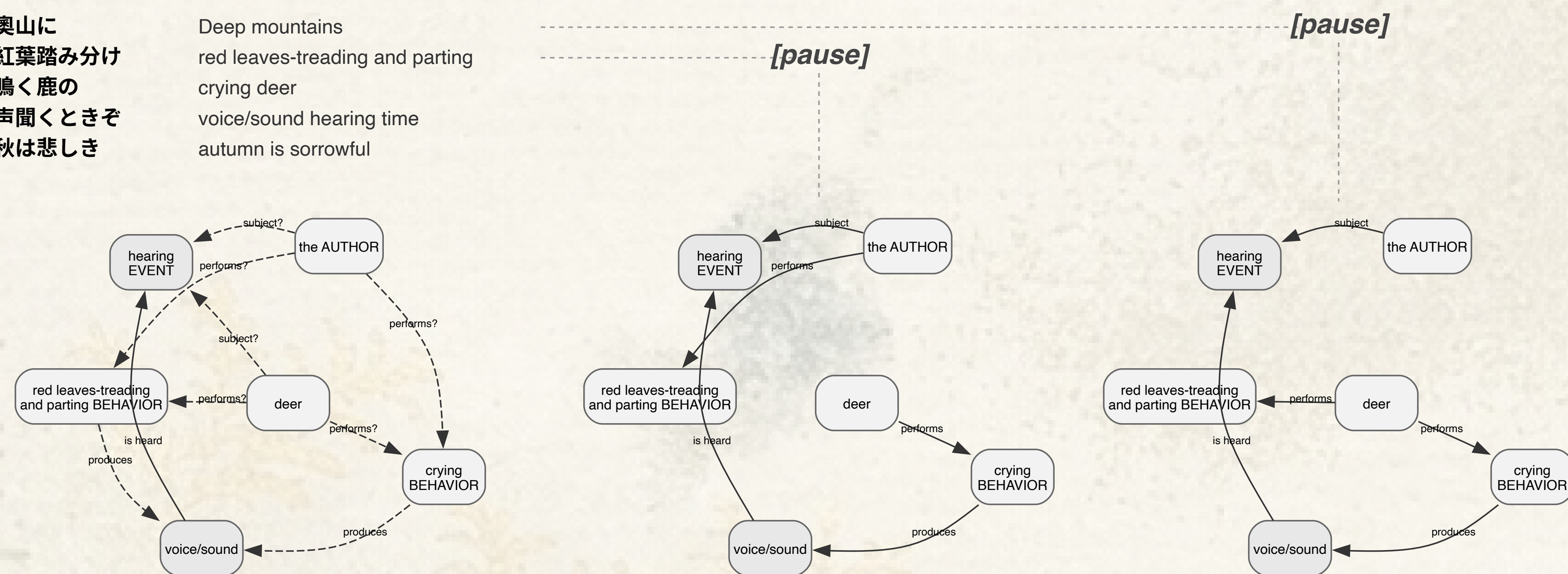


Figure 1: An example for the kugire-based interpretive diversity of poem no. 215

- Kugire has interpretive diversity and reflects discourse-level interpretive diversity
- Lack of structured kugire data
- Kugire data → Ten translators + morphological rules inconsistency

What Does This Work Do?

Classical Japanese poetry, a.k.a. waka, consists of 5/7/5/7/7-syllable phrases. Kugire are grammatical/semantic pauses within the phrases.

Kugire (句切れ) are phrase/clause breaks within waka poems. We try to describe kugire with structured data and provide multiple sources of evidence, covering:

1. Annotating **non-visible** interpretive elements in structured data
2. How annotation preserves the source and reliability of information
3. Interpretive diversity at the discourse level

Why Kugire?

- We perform analysis on interpretive diversity for visible lexical elements in waka
- Analysis on invisible discourse-level elements was not performed yet
- Kugire shows discourse-level interpretability

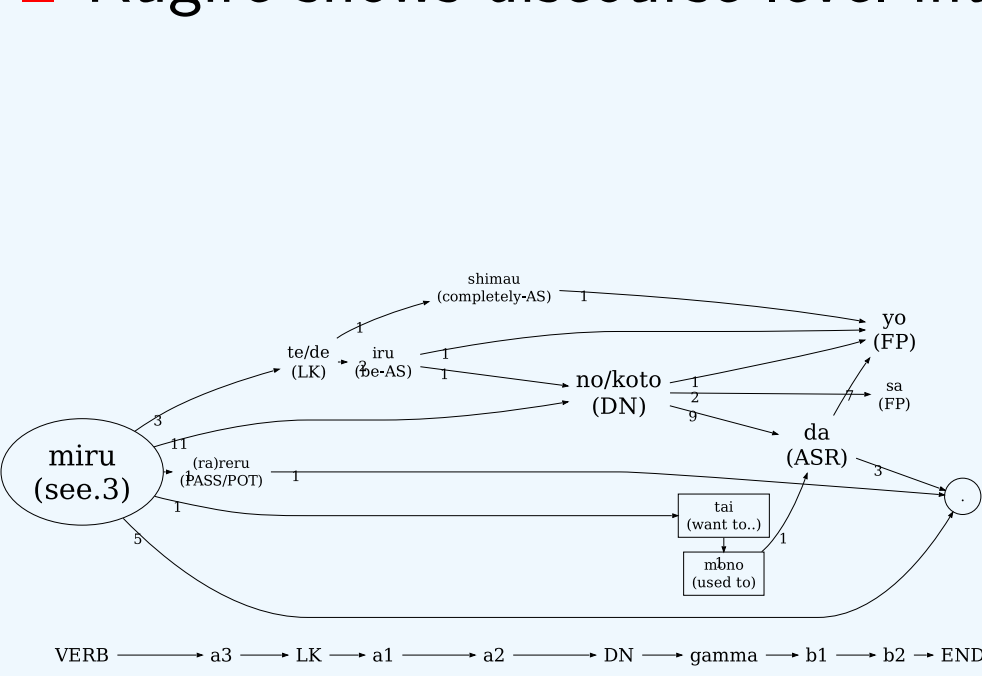


Figure 2: Interpretive diversity of the predicate of *miru* (see)

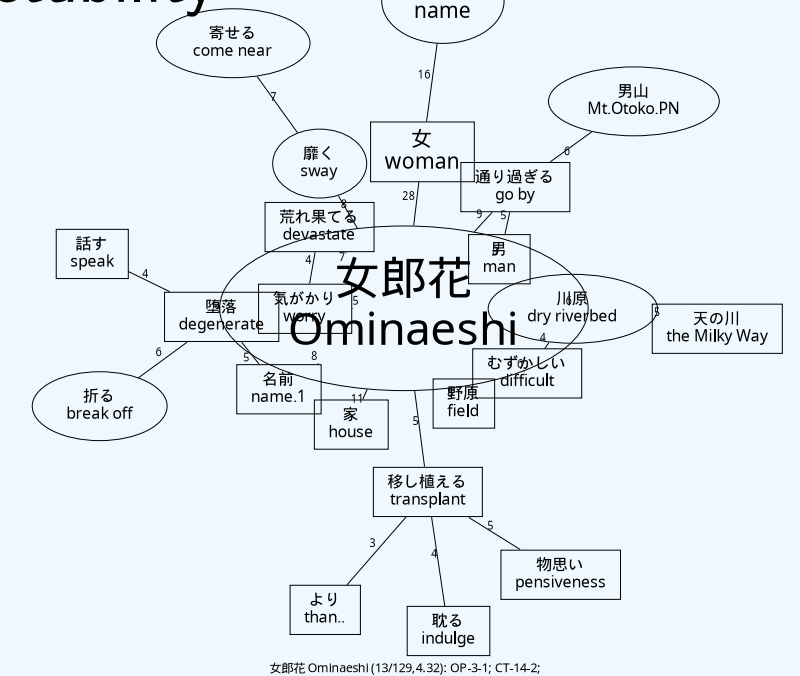


Figure 3: Interpretive diversity of lexical space *ominaeshi* ("maidenflower")

Output data

The output file is *kokin-kugire.xml*. Each kugire candidate is appended as a `<k>` element inside the corresponding `<l>`, keeping at least:

- @n: which segment the kugire falls after
- @source: the judgment source, e.g. morph or kaneko
- @cert: the certainty given by the grammatical rule (high/mid/low)

A single poem can retain multiple `<k>` elements — this preserves both the grammatically visible break point and the interpretive break point evidenced in translation side by side.

`<k>` is a project-internal tag pending a decision on the target TEI encoding. A future revision will replace it with `<anchor>` and `<standOff>`:

```
<seg>年の内に</seg>
<anchor xml:id="#n1_break_after_2"/>
<seg>春はきにけり</seg>
<standOff>
  <interpGrp type="kugire">
    <interp target="#n1_break_after_2"
      source="#morph" cert="high"/>
    <interp target="#n1_break_after_2"
      source="#kaneko"/>
  </interpGrp>
</standOff>
```

Input data

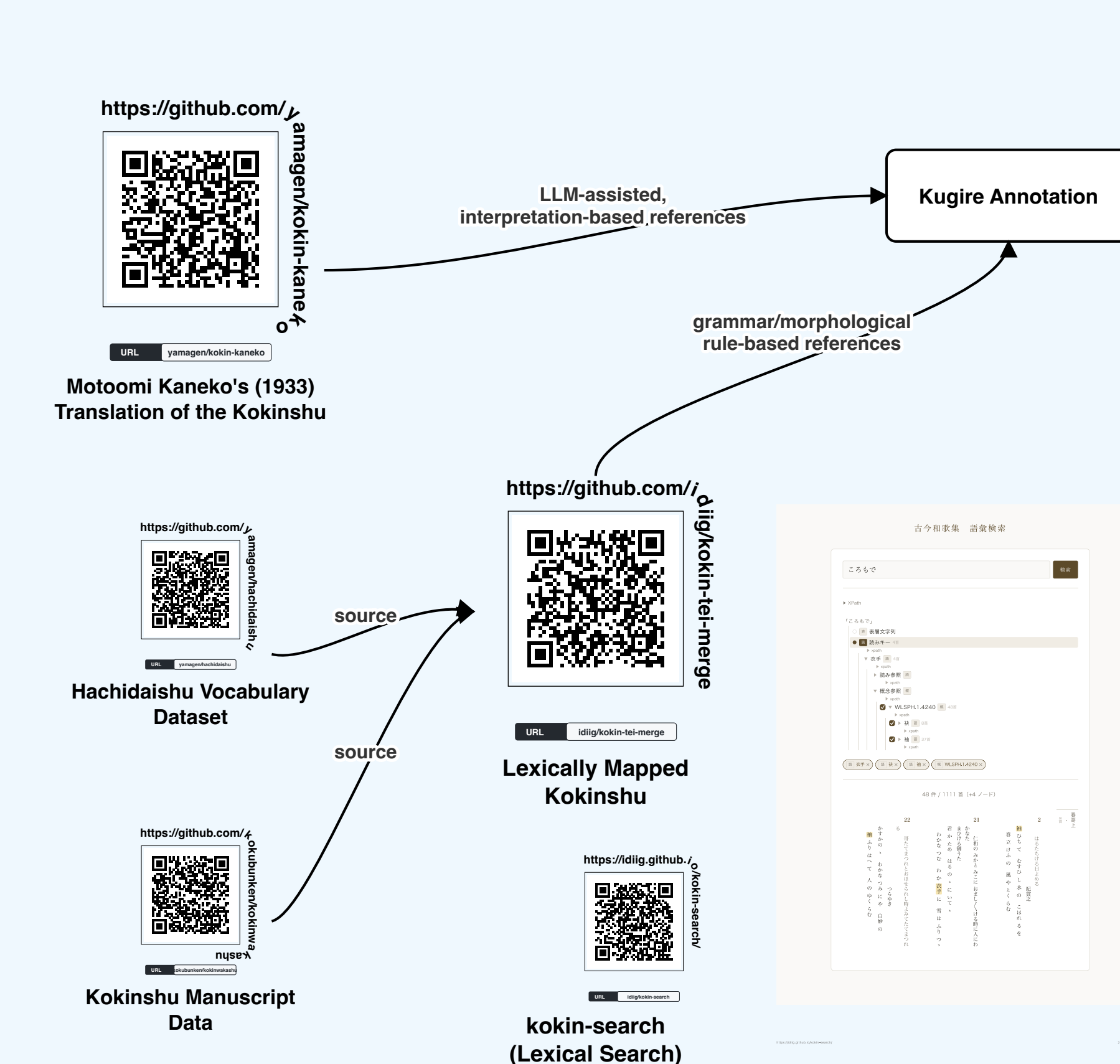


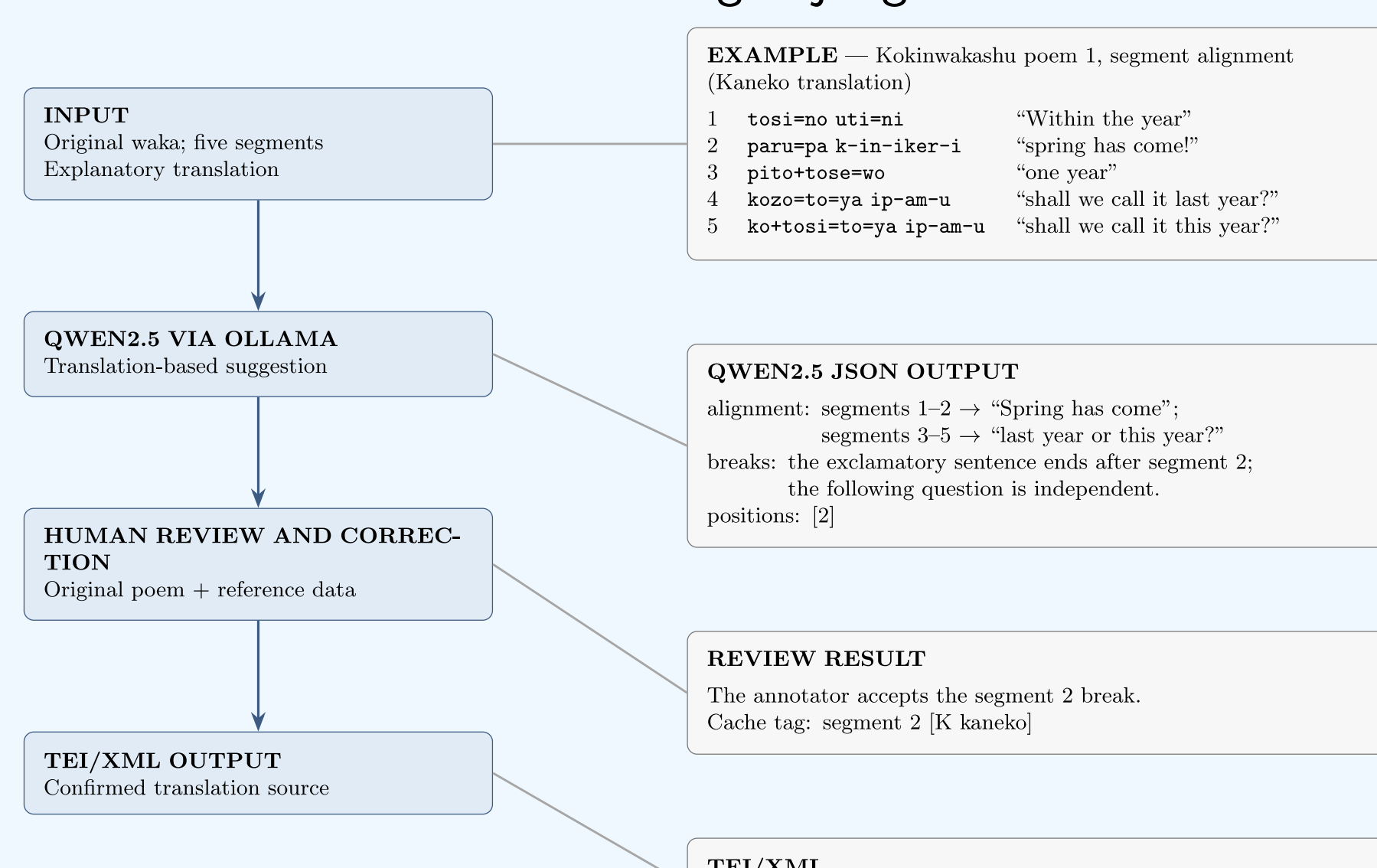
Figure 5: Relationships among the project's data sources and tools: TEI/XML merge, vocabulary datasets, search and network-visualization tools, translation corpus.

1. **Base TEI/XML**: the Kokinwakashu text, five-ku segmentation, author information, textual variants, commentary, and other existing TEI markup
2. **Vocabulary and morphological information**: the Hachidaishu vocabulary list and part-of-speech data; used to infer kugire candidates from inflection forms, sentence-final particles, binding particles, etc.
3. **Modern Japanese translation**: the modern Japanese gloss in Kaneko Motoomi's *Kokinwakashu Hyōshāku*; used to observe how 20th-century commentators construed the waka as modern Japanese discourse

Translation-Based Annotation

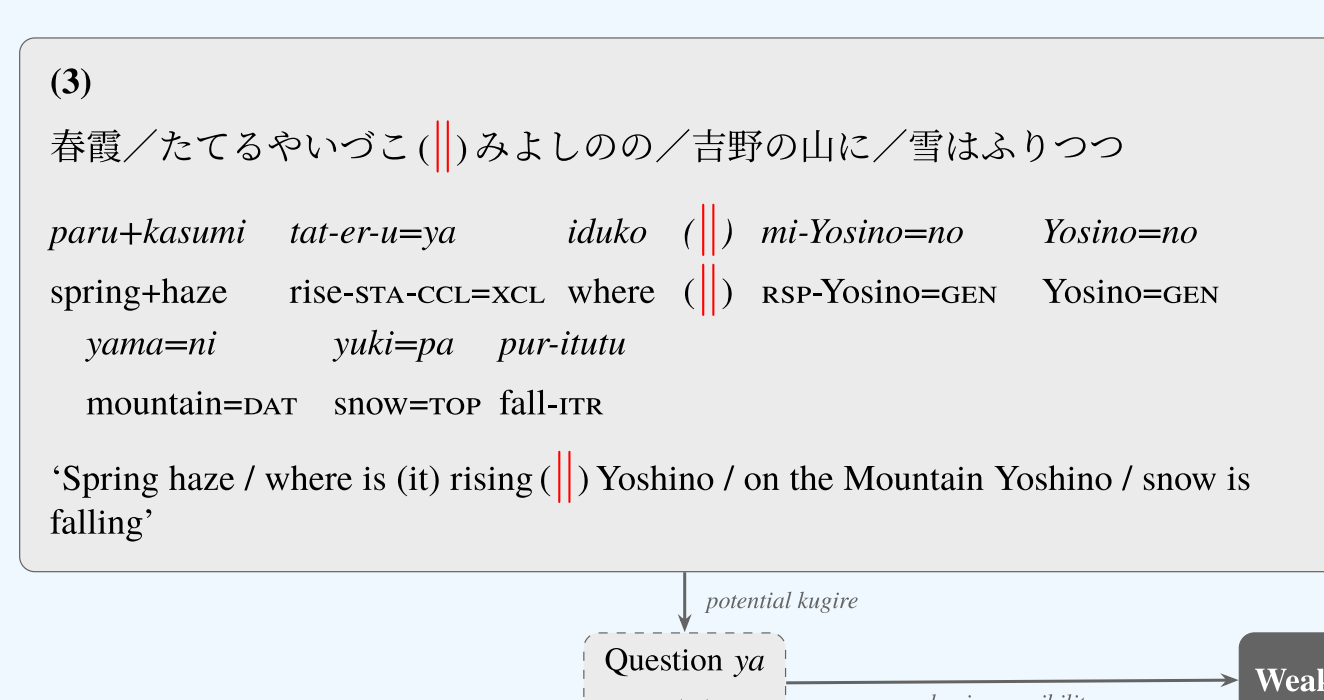
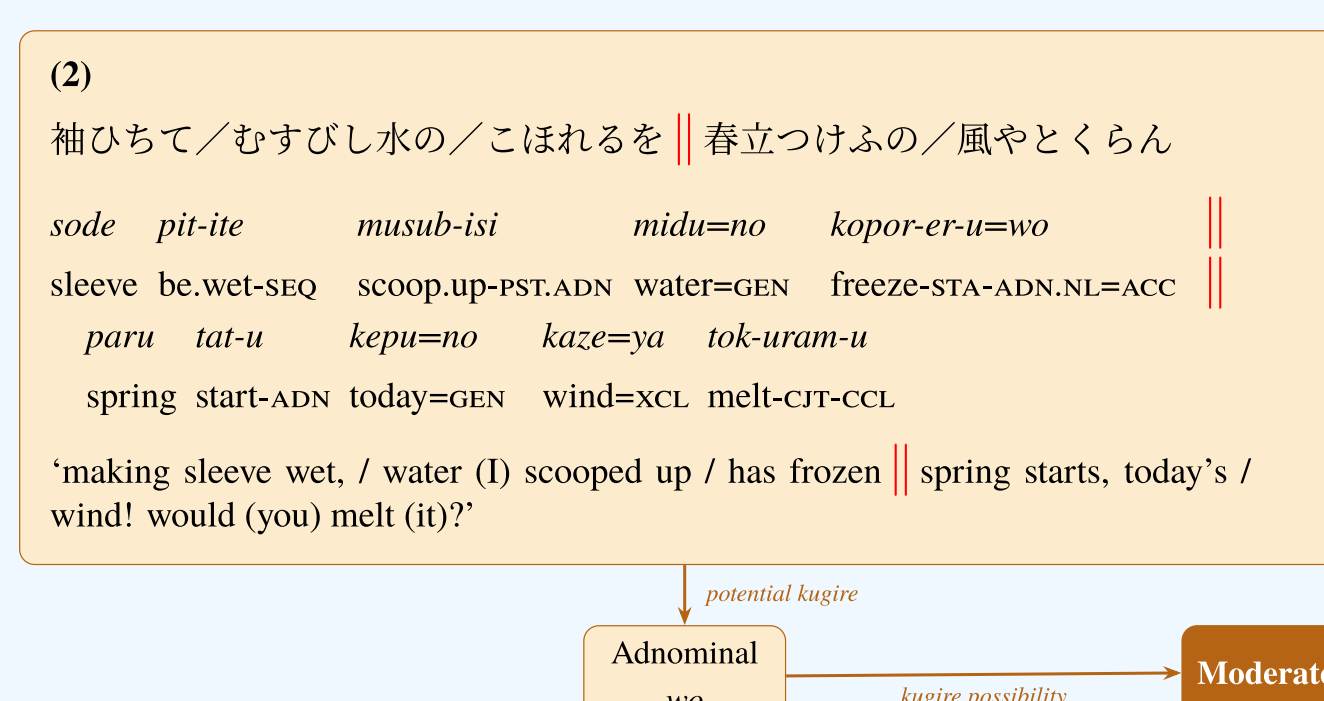
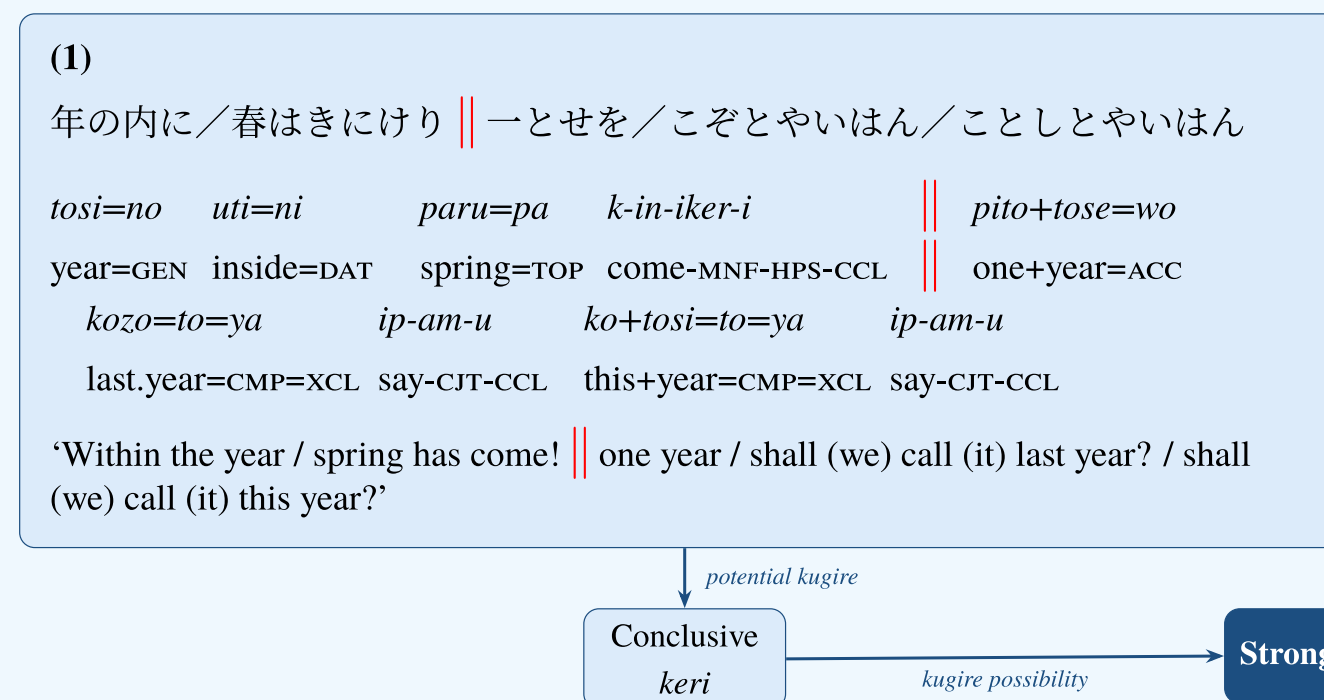
Uses Kaneko's modern Japanese gloss as interpretive evidence. A locally run LLM (qwen2.5 / Ollama) first aligns the poem's five segments with the modern Japanese translation, then finds sentence- or discourse-level breaks in the translation, and finally maps them back onto segment positions in the original poem. This method emphasizes "the places read as broken in interpretive practice."

Human review and TEI output: automatically generated candidates are first written to a per-poem cache file, then checked, added to, or corrected by a human reviewer. The confirmed result is written back to TEI/XML, so the data stays traceable to the evidence behind each kugire judgment.



Morphology-Based Kugire Annotation

Searches the morphological data for positions that may form a kugire. Conclusive forms, imperative forms, realis forms, sentence-final particles, etc. serve as strong candidates; binding particles, kakari-musubi, taigen-dome, etc. serve as moderate-to-weak candidates. This method emphasizes "grammatically observable break points."



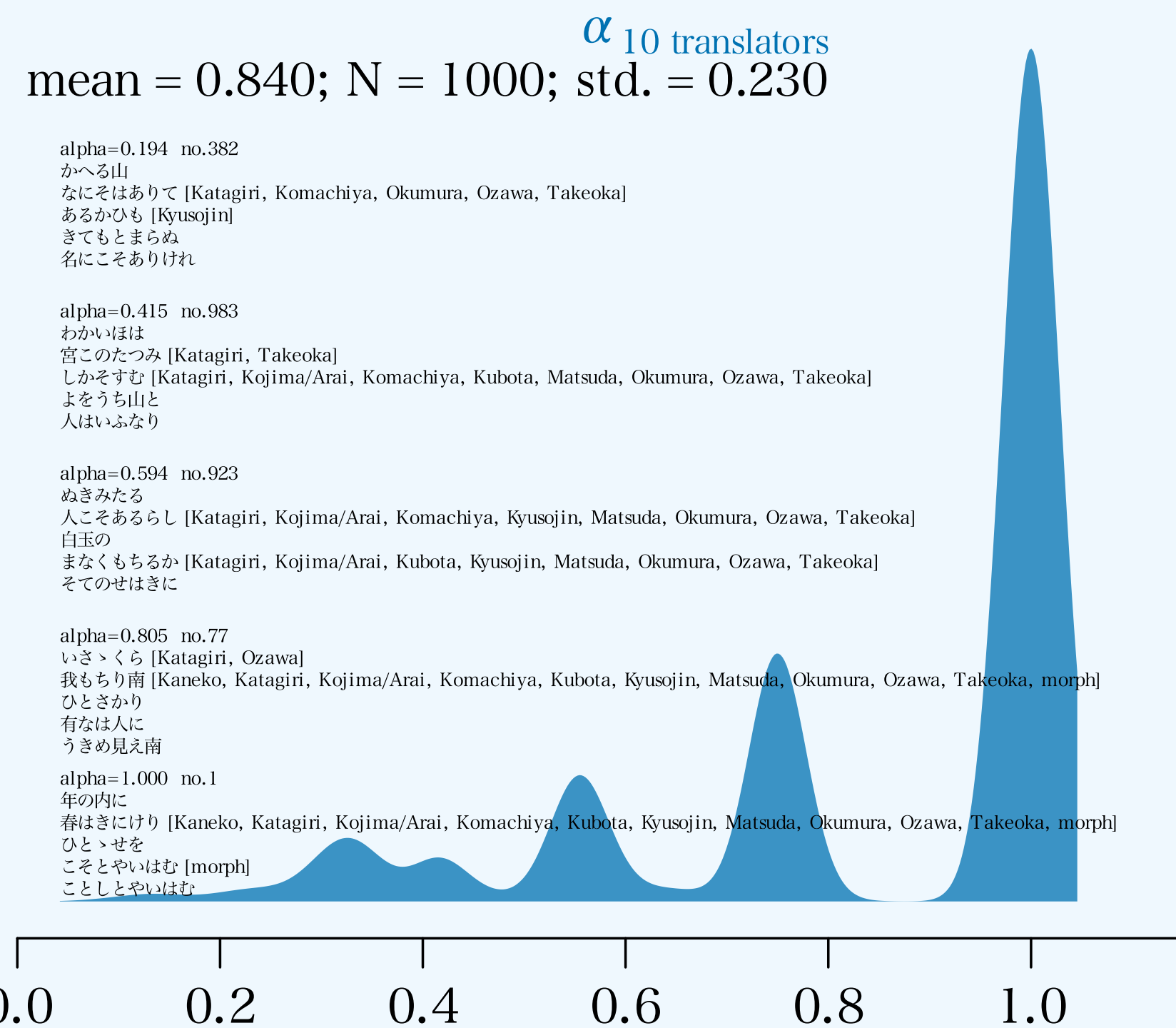
Abbreviations: - = affix boundary; = = clitic boundary; + = compound boundary; ACC = accusative; ADN = adnominal; ADN = adnominal; CCL = conclusive; CCL = conjectural; CMP = complementizer; DAT = dative; GEN = genitive; HPS = hearsay post; ITR = iterative; MAN = manifestive; NL = nominalizer; PST = past; RSP = respect; SEQ = sequential; STA = stative; TOP = topic; XCL = exclamative

Figure 7: Three classes of kugire candidates derived from morphological information: strong, moderate, and weak candidates. This figure illustrates that the rule-based method does not give a single answer, but candidate interpretations with certainty levels.

WIP Results

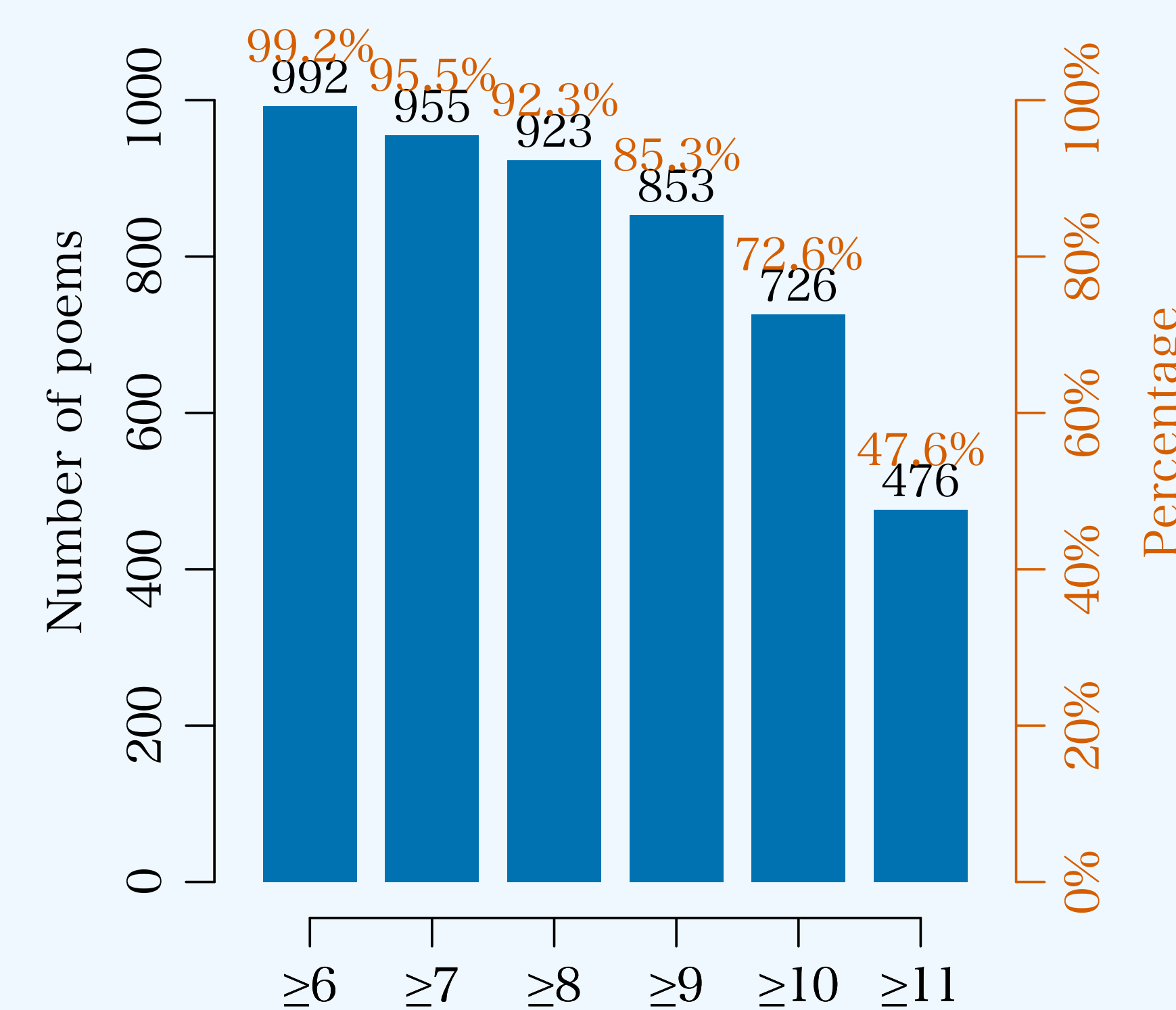
- Broadly similar readings show high agreement, but divergent interpretations always remain

mean = 0.840; N = 1000; std. = 0.230



Krippendorff's α among ten translators

Figure 8: Distribution of agreement among ten translators over four possible kugire positions (Krippendorff's alpha computed per poem).



Minimum number of agreeing judgements

Figure 9: Cumulative inter-judge agreement: number and percentage of poems (out of 1000) where at least N of the 11 judgment sources agree on the kugire position, for N = 6 through 11.